

Yongjin Kim

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RESEARCH INTERESTS	Model-Based GNC; Vision/Optical Navigation		
EDUCATION	Konkuk University <i>B.S. Candidate in Electrical and Electronics Engineering</i>	Mar. 2022 – Feb. 2028 (Expected)	Seoul, Korea
	• GPA: 3.9 / 4.5		
RESEARCH EXPERIENCE	Intelligent Navigation and Control Laboratory , Konkuk University <i>Undergraduate Research Student (Adviser: Professor Sang-Kyung Sung)</i>	Dec. 2025 – Jun. 2026	Seoul, Korea
	• INS/GNSS Integrated Navigation - Implemented and evaluated a 15-state loosely coupled INS/GNSS extended Kalman filter for UGV trajectory estimation. - Analyzed position, velocity, and attitude errors against reference data and documented limitations in initialization, modeling, and tuning.		
PROJECTS	ASEC Rocket Team , Konkuk University <i>Electronics Team Member</i>	Mar. 2025 – Present	Seoul, Korea
	• Hardware-in-the-Loop Avionics Test System (Ongoing) - Developing an STM32F405-based HILS platform for avionics firmware validation. - Designing a packetized interface that replays Simulink-generated sensor data to the flight computer for the 35th NURA rocket.		
	• FPGA-Accelerated Multi-Sensor Navigation System (Ongoing) - Developing an HLS-based 21-state EKF to accelerate real-time navigation processing. - Designing an Arty A7-100T architecture to acquire and synchronize multi-rate IMU, GNSS, barometer, and magnetometer data.		
	• Rocket Telemetry and Ground Control Station - Built a real-time web-based GCS for telemetry visualization during the 34th NURA. - Integrated decoding for IMU, GPS, altitude, attitude, and flight-status data, and verified 447 MHz telemetry reception over approximately 600 m.		
	2026 Engineering Capstone Design Team , Konkuk University <i>Control Software and Electronic Systems Developer</i>	May 2026 – Present	Seoul, Korea
	• Active Suspension System for Stroller Vibration Reduction (Ongoing) - Developing control software, electronic systems, and an IMU-based evaluation workflow for a motor-actuated stroller suspension prototype.		
	Advanced Electrical and Electronics Design & Software Practice , Konkuk University <i>Course Project</i>	Sep. 2026 – Present	Seoul, Korea
	• Vision-Based Landing Target Detection and Tracking (Ongoing) - Developing real-time landing-target detection and relative motion tracking using C++/OpenCV feature matching and optical flow.		

Intelligent IoT Design, Konkuk University
Course Project

Sep. 2026 – Present
Seoul, Korea

- **UWB/IMU Indoor Localization and IoT Monitoring System (Ongoing)**

- Developing a Raspberry Pi 4-based indoor localization system using UWB/IMU sensor fusion, database logging, and web-based monitoring and control.

Fundamentals of Electrical and Electronics Design & Software Practice,

Konkuk University

Mar. 2026 – Jun. 2026

Course Project

Seoul, Korea

- **64-Point FFT Software Optimization on Zynq**

- Completed and benchmarked a 64-point radix-2 DIF FFT in C on the Zybo Z7-20, comparing DFT, C FFT, and ARMv7 assembly across compiler and cache configurations.

- **16-bit Five-Stage Pipelined Tiny CPU**

- Implemented a 16-bit five-stage pipelined CPU in Verilog with arithmetic, memory, and control-flow instructions and multi-stage data forwarding.

AI Accelerator Design, Konkuk University

Mar. 2026 – Jun. 2026

Course Project

Seoul, Korea

- **Tiny Neural Network Accelerator**

- Implemented a Verilog accelerator for FC1-Norm-ReLU-FC2 inference with a reusable 4×4 weight-stationary systolic array, tiled multiplication, and batch-mode control logic.

Microprocessor Applications, Konkuk University

Mar. 2026 – Jun. 2026

Course Project

Seoul, Korea

- **CNN Accelerator Optimization with RISC-V**

- Optimized RISC-V assembly kernels for CNN accelerator data movement, address generation, and memory-access scheduling, with functional validation across 168 convolution cases.

WINEQUEEN, 23rd Embedded Software Contest

Mar. 2025 – Sep. 2025

Vision and Control Software Developer

Seoul, Korea

- **Automated Wine-Bottle Sealing and Opening System**

- Developed YOLOv8 detection and Arduino control software for automated wine sealing and opening.

Digital Communications Laboratory, Konkuk University

Feb. 2025 – Jun. 2025

Undergraduate Project Member

Seoul, Korea

- **Vision-Based Drone Following Project**

- Developed YOLO-based person detection and bounding-box control logic for drone-following experiments.

TRAINING

DYROS Robotics Boot Camp, Seoul National University

Jan. 2026

Participant

Online

- Completed hands-on training in ROS2 communication, MuJoCo simulation, and MoveIt2.

AWARDS

KOSAC President's Award (3rd Place),

2026 KASA Aerospace Academic Competition

Sep. 2026

Team NavBoost, Konkuk University

Korea

1st Place,

35th NURA National University Rocket Launch Competition

Aug. 2026

ASEC Rocket Team, Konkuk University

Korea

	2nd Place, 35 th NURA National University Rocket Academic Conference ASEC Rocket Team, Konkuk University	Jul. 2026 Korea
	Sponsor's Award (4th Place), 34 th NURA National University Rocket Launch Competition ASEC Rocket Team, Konkuk University	Aug. 2025 Korea
LEADERSHIP	BOOT&4DIM, College of Engineering Academic Club, Konkuk University President / Vice President / Executive Board Member	Mar. 2023 – Aug. 2023; Mar. 2025 – Jun. 2026 Seoul, Korea
	- Led an embedded software and hardware prototyping club.	
	Electrical and Electronics Engineering Student Self-Governance Committee, Konkuk University Chair	Jan. 2026 – Aug. 2026 Seoul, Korea
	- Coordinated student governance and communication with the department.	
TECHNICAL SKILLS	Programming Languages: C/C++, Python, MATLAB, Verilog, RISC-V Assembly Hardware Platforms: Arduino, Raspberry Pi, STM32 Tools: Linux, Vivado, KiCad, OrCAD, Fusion 360	
LANGUAGE	Korean: Native	
MILITARY EXPERIENCE	Republic of Korea Army Sergeant, Honorably Discharged	Sep. 2023 – Mar. 2025 Korea